SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection

A city surveillance system must operate under the following conditions:

- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:

Viola-Jones: 78%, low latency
YOLO: 92%, moderate latency
Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms
Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes
Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

5. Depth Estimation Strategy

Constraints:

- Only one camera available
- Moderate depth accuracy acceptable

Options:

Stereo Vision: high accuracy, needs two cameras
Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

6. Retail Motion Tracking Logic

A retail system requires:

- Accurate tracking when crowd density is low
- Stable tracking when density is high

Observations:

Optical Flow: accurate in low density, noisy in high density
Motion Models: stable in high density, less detailed

Compare both and evaluate a suitable approach or combination. Justify your reasoning logically.

7. Defect Detection System Decision

Requirements:

- Detect subtle defects (>95% accuracy)
- Adapt to product variations

Options:

Traditional methods: fast, less adaptive
Deep learning: accurate, requires training

Compare both approaches and evaluate which meets requirements. Justify using performance vs adaptability logic.

8. AR Motion Tracking Constraint Problem

An AR system must:

- Maintain stable tracking (<5% drift)
- Run on mobile hardware

Options:

Optical Flow: detailed but computationally intensive
Motion Estimation: less detailed but efficient

Compare and evaluate which method satisfies both constraints. Justify your decision logically.

9. Mobile Face Detection Trade-off

Constraints:

- Real-time operation required
- Accuracy ≥85%

Options:

Viola-Jones: 80%, fast
Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

10. Animation System Optimization

A game engine requires:

- Realistic motion for cutscenes
- Fast rendering for gameplay

Options:

Motion Models: realistic, computationally heavy
Motion Estimation: efficient, less realistic

Compare both approaches and evaluate how the system should be designed. Justify using scenario-based logical reasoning.

11. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy ≥ 90%
- Latency ≤ 80 ms
- Works under varying lighting

Options:
Method A: 88%, 60 ms
Method B: 91%, 85 ms
Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

12. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy ≥ 85%

Options:
Model A: 83%, 40 ms
Model B: 87%, 65 ms
Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

13. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:
Traditional: 85%, low computation
Deep Learning: 94%, high computation
Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

14. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:
Optical Flow: accurate, slow
Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

15. Face Recognition System Design

Constraints:

- Accuracy ≥ 90%
- Works in low light
- Moderate computation

Options:
Traditional: 82%
Deep Learning: 93%
Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

16. Autonomous Navigation Constraint

System must:

- Accuracy ≥ 95%
- Latency < 70 ms

Options:
Model A: 94%, 60 ms
Model B: 96%, 80 ms
Model C: 95%, 65 ms

Compare and evaluate. Justify your decision.

17. Industrial Inspection System

Requirements:

- Detect defects with ≥ 95% accuracy
- Real-time operation

Options:
Method A: 92%, fast
Method B: 96%, slow
Method C: 95%, moderate

Compare and evaluate the suitable method. Justify your decision.

18. Crowd Monitoring System

Requirements:

- Stable detection in dense crowds
- Moderate accuracy acceptable

Options:
Optical Flow: unstable in dense scenes
Motion Models: stable

Compare and evaluate. Justify your decision.

19. Video Analytics Pipeline

Constraints:

- Accuracy ≥ 90%
- Processing time < 100 ms

Options:
Pipeline A: 88%, 80 ms
Pipeline B: 91%, 120 ms
Pipeline C: 90%, 95 ms

Compare and evaluate the best pipeline. Justify your decision.

20. Smart Parking System

Requirements:

- Detect vehicles accurately
- Low computational cost

Options:
Viola-Jones: low cost, low accuracy
YOLO: moderate cost, high accuracy

Compare and evaluate. Justify your decision.

21. Medical Imaging Analysis

Requirements:

- Very high accuracy (>95%)
- Processing time not critical

Options:
Traditional: 85%
Deep Learning: 97%

Compare and evaluate the suitable approach. Justify your decision.

22. Edge Device Vision System

Constraints:

- Low power
- Real-time performance

Options:
Deep Learning: accurate, heavy
Lightweight Model: moderate accuracy, efficient

Compare and evaluate. Justify logically.

23. Traffic Flow Analysis

Requirements:

- Detect motion patterns
- Robust in crowded scenes

Options:
Optical Flow: detailed, noisy
Motion Estimation: stable

Compare and evaluate. Justify your decision.

24. Retail Analytics System

Requirements:

- Customer tracking
- Moderate accuracy
- Real-time processing

Options:
Method A: accurate, slow
Method B: moderate accuracy, fast

Compare and evaluate. Justify your decision.

25. Object Detection in Foggy Conditions

Requirements:

- Robust under poor visibility
- Accuracy ≥ 85%

Options:
Model A: 80%
Model B: 87%
Model C: 90% but slow

Compare and evaluate. Justify your decision.

26. Video Surveillance Upgrade

Constraints:

- Improve accuracy
- Maintain same latency

Options:
Model A: 85%, 50 ms
Model B: 90%, 50 ms

Compare and evaluate. Justify your decision.

27. Robotics Vision System

Requirements:

- Fast decision-making

- Moderate accuracy

Options:
Deep Learning: slow
Traditional: fast

Compare and evaluate. Justify your decision.

28. Smart Home Security

Requirements:

- Detect intrusions accurately
- Low computation

Options:
YOLO: accurate
Viola-Jones: efficient

Compare and evaluate. Justify your decision.

29. Autonomous Drone Tracking

Constraints:

- Real-time (<40 ms)
- Accuracy ≥ 85%

Options:
Model A: 83%, 30 ms
Model B: 86%, 50 ms
Model C: 88%, 38 ms

Compare and evaluate. Justify your decision.

30. Motion Animation System

Requirements:

- Smooth motion
- Real-time rendering

Options:
Motion Models: realistic, slow
Motion Estimation: fast, less realistic

Compare and evaluate design approach. Justify your decision.

Code of Conduct:

I _____BHAVYA PS(192524017)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Bhavya ps

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understand ing of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understandi ng with minor gaps in requirement s	Basic understandi ng; some requirement s unclear	Poor understandin g; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. Smart Surveillance System Selection

Given:

- Latency > 100 ms → system fails. Accuracy < 85% → unacceptable. Hardware
- supports only moderate computation.

Model	**Accuracy**	**Latency / Computation**
Viola-Jones	78%	Low latency
YOLO	92%	Moderate
Deep Learning	95%	High

Comparison:

- Viola-Jones has low latency and low computation, but its accuracy is
- only 78%, which is below the required 85%.
- YOLO has 92% accuracy and requires moderate computation, making it
- suitable for the available hardware.
- Deep Learning has the highest accuracy of 95%, but its high latency
- and computation make it unsuitable for the given constraints.

Analysis:

- The system must satisfy all the constraints, not just one.
- Viola-Jones fails because its accuracy is below 85%.
- Deep Learning provides excellent accuracy, but its high latency and
- computation requirements make it unsuitable.
- YOLO provides a practical balance between accuracy and system
- performance.

Decision:

YOLO

Conclusion:

YOLO is the best choice because it satisfies the accuracy requirement and is compatible with moderate hardware computation. Viola-Jones is rejected due to low accuracy, while Deep Learning is rejected because of its high latency and computation requirements.

2. Autonomous Vehicle Detection Logic

Given:

- Safety requires accuracy > 90%. Response time must be < 50 ms.

Method	Accuracy	Response Time
A	88%	30 ms
B	93%	70 ms
C	91%	45 ms

Comparison:

- Method A meets the response-time requirement but fails the accuracy
- requirement.
- Method B meets the accuracy requirement but fails the response-time
- requirement.
- Method C meets both accuracy and response-time requirements.

Analysis:

- For an autonomous vehicle, both accuracy and response time are
- critical for safety.
- A method with high accuracy but slow response can still be dangerous
- because the vehicle may react too late.
- Method C provides 91% accuracy and 45 ms response time, satisfying
- both constraints.

Decision:

Method C

Conclusion:

Method C is the only method that satisfies both safety constraints. Therefore, it is the most suitable option for autonomous vehicle detection.

3. Motion Analysis Decision Problem

Given:

- Accurate motion direction detection is required. High robustness in
- dense traffic is required.

Approach	Advantage	Limitation
Optical Flow	Accurate motion information	Noisy in dense scenes
Motion Estimation	Stable and robust	Less detailed

Comparison:

- Optical Flow provides detailed and accurate motion information, but
- its vectors may become noisy when many vehicles are moving close to
- each other.
- Motion Estimation provides more stable motion information and performs
- better when the scene becomes complex, but it may not capture small or
- detailed movements.

Analysis:

- In dense traffic, many vehicles move simultaneously, which can make
- Optical Flow sensitive to noise and incorrect motion vectors.
- Motion Estimation is more stable under these conditions.
- Although it provides less detailed information, its stability is more
- important when the system must continuously track traffic movement in
- a crowded environment.

Decision:

Motion Estimation

Conclusion:

For high-density traffic, Motion Estimation is more suitable because stability and robustness are more important than very detailed motion information. Optical Flow can be preferred in less crowded scenes where detailed motion detection is required.

4. Background Modeling Evaluation

Given:

- Static scenes: 90% accuracy. Dynamic scenes: 65% accuracy. Requirement:
- System must maintain ≥ 80% accuracy across all scenarios.

Scene Type	GMM Accuracy
Static scenes	90%
Dynamic scenes	65%

Comparison:

- In static scenes, GMM achieves 90% accuracy, which satisfies the
- requirement.
- In dynamic scenes, GMM achieves only 65% accuracy, which is below the
- required 80%.

Analysis:

- The system requirement applies to all scenarios, not only static
- scenes.
- Although GMM performs well in static environments, its performance
- drops significantly in dynamic environments because background changes
- can be incorrectly detected as moving objects.
- Since the dynamic-scene accuracy is only 65%, GMM alone cannot
- guarantee the required 80% accuracy.

Decision:

GMM alone should not be used.

Conclusion:

GMM alone does not meet the requirement because its accuracy falls to 65% in dynamic scenes. A better solution is to combine GMM with adaptive background updating, motion filtering, Optical Flow, noise reduction, or shadow removal.

5. Depth Estimation Strategy

Given:

• Only one camera is available. Moderate depth accuracy is acceptable.

Method	**Accuracy**	**Camera Requirement**
Stereo Vision	High	Two cameras
Structure from Motion	Moderate	One camera

Comparison:

• Stereo Vision provides high depth accuracy but requires two cameras.
• Structure from Motion provides moderate depth accuracy and works with
• a single camera.

Analysis:

• The most important constraint is that only one camera is available.
• Stereo Vision cannot directly satisfy this requirement because it
• depends on two different camera views to calculate depth.
• Structure from Motion is more suitable because it estimates depth
• using the movement of a single camera across multiple frames.
• Since moderate accuracy is acceptable, the lower accuracy compared
• with Stereo Vision is not a major problem.

Decision:

Structure from Motion

Conclusion:

Structure from Motion is the most feasible option because it satisfies both constraints: single-camera operation and acceptable moderate depth accuracy. Stereo Vision provides better accuracy but cannot be selected because it requires two cameras.